# Topic 7.1 Chromosomes, genes and DNA

## Exercise 7.1 Chromosomes, genes and DNA

**1** A: nucleus; B: cytoplasm

**2** X written in the nucleus of each cell.

**3** Look for a sentence that includes correct information about each term. Example sentences are:

Chromosomes are found in the nucleus of every cell / Chromosomes are made of DNA.

Genes are found on chromosomes / Genes contain information that is inherited from an organism's parents.

DNA is the substance that chromosomes and genes are made of.

**4** The DNA in a cell contains instructions for making a cell and a whole organism. If any of these instructions are missing or damaged then the cell or the organism cannot be correctly formed. So when a cell divides it is essential that each of the new cells gets a full set of all the DNA in the parent cell.

# Topic 7.2 Gametes and inheritance

## Exercise 7.2 Egg cells and sperm cells

1 Egg cells and sperm cells are specialised cells called **gametes**.

Egg cells are **female** and sperm cells are **male**.

All **sperm** cells contain one X chromosome, but **egg** cells can contain an X chromosome or a Y chromosome.

A sperm cell can join with an egg cell in a process called **fertilisation**.

**2** For example:

| Egg cell | Sperm cell |
|---|---|
| has food reserves in the cytoplasm | does not have food reserves in the cytoplasm |
| round in shape | elongated |
| no tail | has a tail |
| has an extra layer round the outside of the cell | does not have layer round the outside of the cell |
| larger | smaller |
| has a cell membrane | has a cell membrane |
| has a nucleus | has a nucleus |
| has cytoplasm | has cytoplasm |

**3** Accept any statement that makes clear what the difference is and that states a sensible reason for the difference. For example:

Sperm cells have a tail to swim to the egg cell, but egg cells do not need to swim.

Sperm cells are elongated to make it easier for them to swim, but egg cells do not move so they can just be round.